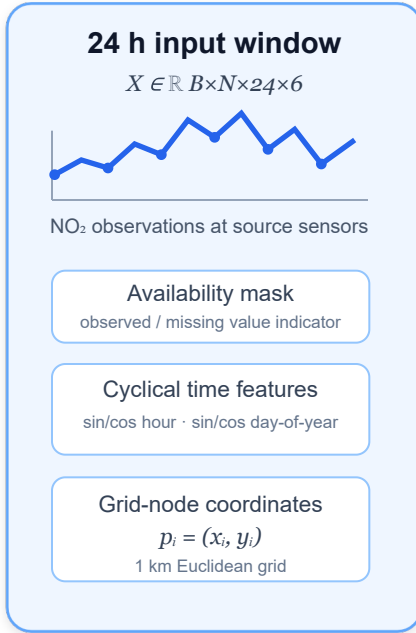


Sensor-to-Grid Spatio-Temporal GNN

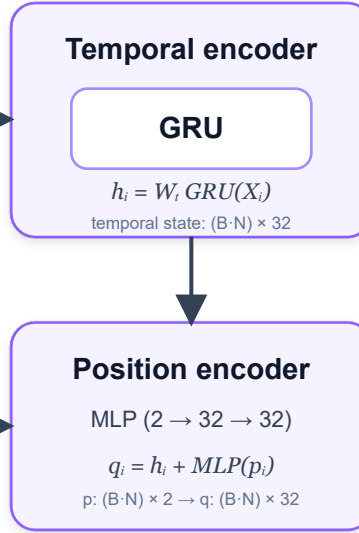
Global learned sensor influence followed by local spatial refinement

INPUT

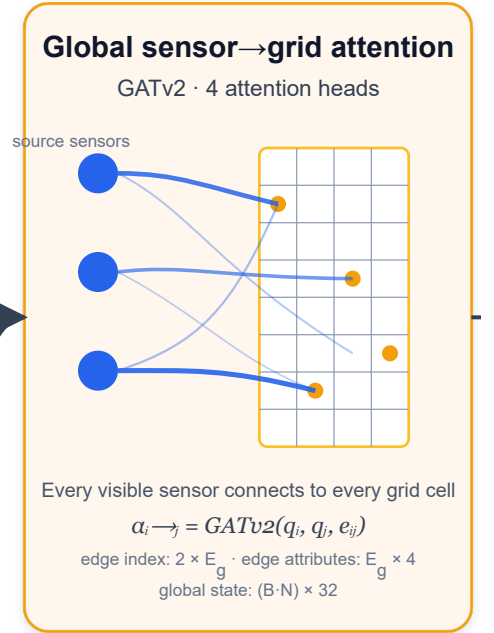


Observed AQ is exposed only at context sensors

ENCODING



GLOBAL PROPAGATION



Tensor notation: B batch $\cdot$ N grid nodes $\cdot$ $T=24$ hours $\cdot$ $H=32$ hidden

Input channels $C=6$: NO₂ value, availability, $\sin/\cos$ hour, $\sin/\cos$ day-of-year

Residual connections $\cdot$ ELU $\cdot$ LayerNorm $\cdot$ Dropout (0.1) $\cdot$ masked MAE at held